

GOVERNMENT WELFARE ADVISORY SYSTEM



A PROJECT REPORT

Submitted by

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In partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

GOVERNMENT COLLEGE OF ENGINEERING,

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ANNA UNIVERSITY: CHENNAI 600025

MARCH 2026

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BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

We sincerely express our whole-hearted thanks to our Principal **Dr. A. SARADHA M.E., Ph.D.**, Government College of Engineering, Erode for his constant encouragement and invaluable support during the course of this project.

We owe our sincere thanks to **Dr. A. KAVITHA, M.E., Ph.D., Professor and Head of the Department**, Department of Computer Science and Engineering, Government College of Engineering, Erode, for providing every essential facility required for completing this project.

We sincerely thank our guide **Prof. S. VIJAYALAKSHMI M.E., Assistant Professor**, Department of Computer Science and Engineering, Government College of Engineering, Erode, for her valuable help and guidance throughout the project.

We sincerely thank our Coordinator **Dr. S. KALAIVANI, M.E., Ph.D., Assistant Professor**, Department of Computer Science and Engineering, Government College of Engineering, Erode, for her valuable help and guidance throughout the project.

We wish to express our sincere thanks to all the staff members of the Department of Computer Science and Engineering for their valuable suggestions and guidance rendered to us throughout the project.

Above all, we are grateful to all our classmates and friends for their friendly cooperation and their exhilarating company throughout the course of this endeavour.

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ABSTRACT

The **Government Welfare Advisory System** is a web-based application designed to help citizens easily discover and apply for government welfare schemes that they are eligible for. In many cases, people are unaware of the benefits and schemes provided by the government due to lack of proper information, complex eligibility rules, or difficulty navigating multiple government websites. This project aims to simplify the process by creating a centralized platform where users can check their eligibility and receive recommendations for relevant welfare schemes.

The system uses a full-stack architecture consisting of a React-based frontend for the user interface and a Spring Boot backend for handling application logic and data processing. A MySQL database is used to store information related to users, schemes, applications, and eligibility history. The platform supports multiple roles including citizens, government officers, and administrators, each with specific functionalities. Citizens can check their eligibility, view recommended schemes, submit applications, and track their application status. Officers can review submitted applications and approve or reject them based on verification. Administrators manage schemes, users, and overall system operations.

To improve the recommendation process, the system integrates a machine learning model that predicts the eligibility score of a user based on factors such as age, gender, income, occupation, and social category. This predictive approach reduces the time required to manually search through multiple government portals.

Overall, the Government Welfare Advisory System provides a more organized and accessible way for citizens to explore government benefits while also supporting authorities in efficiently managing welfare scheme applications.

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
AI	Artificial Intelligence
API	Application Programming Interface
BE	Bachelor of Engineering
CSE	Computer Science and Engineering
CSS	Cascading Style Sheets
DB	Database
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
JPA	Java Persistence API
JSON	JavaScript Object Notation
ML	Machine Learning
MVC	Model View Controller
MySQL	My Structured Query Language
npm	Node Package Manager
REST	Representational State Transfer
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1

INTRODUCTION

Government welfare schemes play a crucial role in improving the socio-economic conditions of citizens by providing financial assistance, healthcare support, education benefits, housing facilities, and other essential services. These schemes are introduced by governments to support different sections of society such as low-income families, farmers, students, senior citizens, and minority communities. The primary objective of these programs is to reduce poverty, promote equality, and ensure that essential resources reach the people who need them the most. However, despite the availability of many welfare schemes, a large number of eligible citizens are unable to take advantage of them.

One of the major reasons for this issue is the lack of awareness and accessibility. Information about government schemes is often spread across multiple websites and portals, making it difficult for citizens to find relevant programs. Many people are also unaware of the eligibility criteria or the application procedures required to receive these benefits. As a result, individuals who qualify for these schemes may never apply for them simply because they do not know they exist or do not understand how to access them. This problem becomes even more significant for users with limited digital literacy.

To address these challenges, the **Government Welfare Advisory System** has been developed as a centralized digital platform that helps citizens easily discover and apply for welfare schemes. The main aim of this system is to simplify the process of identifying suitable schemes based on a user's personal and socio-economic details. By entering information such as age, income level, occupation, gender, and category, users can receive recommendations for government schemes that match their eligibility profile. This reduces the need for

manual searching across multiple websites and improves the chances of citizens accessing the benefits they deserve.

The system is designed using a modern full-stack architecture that combines both frontend and backend technologies. The user interface is developed using React and TypeScript, providing a responsive and user-friendly experience. The backend is implemented using Spring Boot, which handles business logic, user authentication, and communication with the database. A MySQL database is used to store information related to users, welfare schemes, applications, and eligibility history.

In addition to these technologies, the system integrates a machine learning model that helps predict a user's eligibility score based on their provided information. This predictive approach improves the accuracy of scheme recommendations and allows users to quickly identify the most relevant welfare programs.

Overall, the Government Welfare Advisory System aims to bridge the gap between citizens and government welfare programs by providing a simple, accessible, and efficient platform for discovering and applying for government benefits.

1.1 Problem Statement

How can we develop a system that helps citizens easily discover government welfare schemes and determine their eligibility, while simplifying the application and management process for both citizens and government authorities?

- Many citizens are unaware of the welfare schemes provided by the government due to scattered information across multiple websites and portals.

- People often struggle to understand the eligibility criteria of different schemes, which vary based on income, age, occupation, and social category.
- Existing government portals mostly provide only scheme listings, requiring users to manually search and analyse whether they qualify for each scheme.
- Citizens may apply for schemes without knowing if they are eligible, which increases the number of invalid applications submitted to the system.
- The absence of personalized recommendations makes it difficult for users to identify schemes that match their personal and socio-economic conditions.
- Government officers have to manually review a large number of applications, which increases workload and slows down the approval process.
- Managing application records, eligibility checks, and scheme data across multiple platforms can create inefficiencies in the overall system.
- Citizens often find it difficult to track the status of their submitted applications, which reduces transparency in the welfare distribution process.
- Lack of a centralized digital platform makes it harder for both citizens and administrators to efficiently access and manage welfare scheme information.
- There is a need for an intelligent, centralized system that can recommend suitable welfare schemes, simplify application processes, and support efficient management by government authorities.

1.2 Motivation of Welfare System

Government welfare schemes are created to support citizens by providing financial assistance, education benefits, healthcare support, and other essential services. These schemes are especially important for economically weaker sections of society who rely on government assistance to improve their quality of life. However, in many cases, eligible individuals do not benefit from these programs because they are unaware of the available schemes or find it difficult to understand the eligibility criteria and application procedures. This gap between the availability of welfare programs and citizens' access to them highlights the need for a more organized and user-friendly system.

With the increasing use of digital technology, it is possible to simplify the way citizens interact with government services. A centralized digital platform can help users easily explore welfare schemes, check their eligibility, and apply for suitable programs without navigating through multiple websites. Integrating modern technologies such as web applications, databases, and machine learning can further improve the efficiency of this process by providing personalized recommendations based on user information. This motivates the development of the **Government Welfare Advisory System**, which aims to bridge the gap between citizens and welfare schemes while making the overall process more transparent and accessible.

Motivating Factors:

- Many citizens are unaware of government welfare schemes that they are eligible for.
- Information about schemes is often distributed across multiple platforms, making it difficult to access.

- Eligibility criteria are complex and difficult for users to interpret without guidance.
- Citizens spend significant time searching for schemes manually.
- Government officers receive many applications that do not meet eligibility requirements.
- There is a need for a system that can recommend suitable schemes based on user data.
- A centralized platform can simplify the process of checking eligibility and applying for schemes.
- Technology can improve transparency and efficiency in welfare scheme distribution.
- Citizens should be able to easily track the status of their applications.
- A smart digital solution can ensure that government benefits reach the people who truly need them.

1.3 Project Scope

The **Government Welfare Advisory System** is designed to provide a centralized platform that helps citizens easily discover government welfare schemes and determine their eligibility. Many people are unable to benefit from available schemes due to lack of awareness and difficulty in understanding eligibility requirements. This system simplifies the process by allowing users to enter their personal and socio-economic information and receive recommendations for schemes that match their eligibility criteria.

The project also supports different user roles such as citizens, officers, and administrators to ensure efficient management of welfare schemes and applications. Citizens can explore schemes, apply for benefits, and track application status, while officers review and process applications. Administrators

manage scheme information and system users. By integrating web technologies and machine learning, the system improves accessibility, transparency, and efficiency in welfare scheme distribution.

Scope of the Project:

- Development of a centralized web platform for accessing government welfare schemes.
- Allowing citizens to register, log in, and manage their personal profiles.
- Providing eligibility checks based on user information such as income, age, and occupation.
- Recommending suitable government schemes using a machine learning model.
- Enabling users to apply for welfare schemes through the system.
- Allowing users to track the status of their applications.
- Providing officers with a dashboard to review and manage applications.
- Enabling administrators to manage schemes, users, and officer assignments.
- Maintaining records of applications and eligibility history in a structured database.
- Improving transparency, accessibility, and efficiency in welfare scheme management.

1.4 Project Objective

The primary objective of the **Government Welfare Advisory System** is to simplify the process of discovering and accessing government welfare schemes for citizens. Many individuals struggle to identify the schemes they are eligible for due to scattered information and complex eligibility criteria. This project aims

to provide a centralized digital platform where users can easily check their eligibility, explore relevant welfare programs, and apply for schemes through a simple and user-friendly interface.

Another important objective of this project is to improve the efficiency of welfare scheme management for government authorities. By providing dedicated dashboards for officers and administrators, the system helps streamline the application review process and maintain organized records of schemes, users, and applications. The integration of machine learning for eligibility prediction further enhances the system by providing personalized recommendations and reducing the time required for manual scheme searches.

Specific Objectives:

- To develop a centralized platform for accessing information about government welfare schemes.
- To allow citizens to register, log in, and manage their profiles within the system.
- To provide an eligibility checking system based on user details such as income, age, and occupation.
- To recommend suitable welfare schemes using a machine learning model.
- To enable citizens to apply for schemes directly through the platform.
- To allow users to track the status of their submitted applications.
- To provide officers with tools to review, approve, or reject applications.
- To allow administrators to manage schemes, users, and system operations.
- To maintain a structured database for storing scheme information, user details, and application records.
- To improve transparency and efficiency in the welfare scheme application process.

CHAPTER 2

LITERATURE REVIEW

2.1 Technologies and Approaches

Digital Platforms for Government Welfare Services

Government welfare schemes are designed to support citizens by providing financial assistance, healthcare benefits, education support, and social security services. However, several studies indicate that many eligible citizens fail to benefit from these programs due to lack of awareness and difficulty accessing information. Traditional government portals often contain large amounts of scheme-related information, but the data is usually distributed across multiple websites. This fragmented structure makes it difficult for citizens to identify suitable schemes quickly.

Researchers suggest that centralized digital platforms can significantly improve the accessibility of welfare schemes. By integrating scheme details, eligibility conditions, and application processes into a single platform, users can easily explore available programs and understand the requirements. Such platforms not only improve accessibility but also simplify the process of applying for government benefits.

Welfare Scheme Recommendation Systems

Recent research has focused on the development of intelligent recommendation systems that help citizens identify government schemes that match their personal and socio-economic conditions. Sharma et al. [1] proposed a system called **GovGenius**, which provides a web-based platform for recommending government welfare schemes to citizens. The system collects user information such as age, income, occupation, and social category and compares it with scheme eligibility criteria to generate recommendations.

The research highlights that many citizens remain unaware of welfare schemes due to the absence of structured advisory systems. By providing personalized recommendations, the system helps users easily identify relevant programs without manually searching through multiple government websites. This approach improves accessibility and encourages greater participation in government welfare initiatives.

Artificial Intelligence in Welfare Management Systems

Artificial intelligence has become an important tool in improving public service delivery. Singh et al. [2] proposed an **AI-driven welfare scheme assistance system** that uses machine learning techniques to analyze user profiles and recommend suitable government schemes. The system processes demographic attributes such as age, education level, income, and occupation to determine eligibility for different welfare programs.

The study highlights that traditional manual approaches for identifying eligible beneficiaries can be inefficient and time-consuming. AI-based systems can automate the process of analyzing user data and matching it with scheme criteria. This reduces the time required to identify relevant schemes and improves the efficiency of government service delivery.

Machine Learning for Beneficiary Identification

Another area of research focuses on the use of machine learning algorithms to identify eligible beneficiaries for welfare schemes. Reddy et al. [3] proposed a **smart beneficiary scheme mapping system** that analyzes socio-economic data such as income level, employment status, and demographic characteristics to predict eligibility for welfare programs.

The study demonstrates that machine learning algorithms such as Decision Trees and Random Forest can be used to classify individuals based on their eligibility for government benefits. By using predictive models, governments can

improve the accuracy of beneficiary identification and reduce errors in the welfare distribution process. This approach also helps reduce administrative workload by automatically filtering eligible applicants.

Summary of Literature Findings

From the reviewed literature, it is evident that digital platforms combined with machine learning techniques can significantly improve the accessibility and efficiency of welfare scheme management systems. Intelligent recommendation systems help citizens identify suitable welfare programs based on their personal information, while predictive models assist governments in accurately identifying eligible beneficiaries.

The proposed **Government Welfare Advisory System** builds upon these concepts by integrating a centralized web platform with a machine learning-based eligibility prediction model. The system allows citizens to enter their personal and socio-economic information, automatically evaluates eligibility conditions, and recommends relevant welfare schemes. This approach aims to simplify scheme discovery, improve transparency, and ensure that welfare benefits reach the intended beneficiaries more efficiently.

References (IEEE)

- [1] A. Sharma, R. Kumar, and S. Patel, “GovGenius – Government Schemes Recommendation Web Application,” ResearchGate, 2024.
- [2] P. Singh, A. Verma, and K. Gupta, “AI-Driven Welfare Scheme Assistance and Management System,” *International Journal of Research Publication and Reviews (IJRPR)*, vol. 6, no. 4, pp. 1203–1208, 2025.
- [3] M. Reddy, S. Rao, and V. Kumar, “Smart Beneficiary Scheme Mapping System Using Machine Learning,” *International Journal of Creative Research Thoughts (IJCRT)*, vol. 13, no. 2, pp. 456–461, 2025.

2.2 Existing System

Government Welfare Scheme Portals

Currently, government welfare schemes are mainly accessed through official government portals and websites. These platforms provide information about various schemes, including eligibility criteria, benefits, and application procedures. Citizens who wish to apply for welfare programs must visit these websites, search for relevant schemes, and manually read the details to determine whether they qualify. Some portals also allow users to submit applications online and track the status of their requests.

Manual Scheme Identification Process

Although these portals provide access to important information, they mostly function as information repositories rather than intelligent advisory systems. Users are often presented with long lists of schemes without personalized recommendations based on their profile. As a result, citizens must spend considerable time analyzing scheme requirements and comparing them with their own details to determine eligibility. This process can be confusing, especially for individuals with limited technical knowledge or digital experience.

Administrative Verification Process

In many existing systems, government authorities must manually review and verify the applications submitted by citizens. Officers check each application to ensure that the applicant meets the eligibility criteria of the selected scheme. This manual verification process increases administrative workload and may delay the approval or rejection of applications. Therefore, while current systems provide basic access to welfare scheme information, they do not fully address the challenges related to scheme discovery, eligibility analysis, and efficient application management.

2.2.1 Limitations of Existing System

- Welfare scheme information is spread across multiple government websites and portals.
- Citizens must manually search for schemes and check eligibility requirements.
- Existing systems do not provide personalized scheme recommendations.
- Understanding eligibility criteria can be difficult for many users.
- The process of identifying suitable schemes is time-consuming.
- Many eligible citizens remain unaware of the schemes they qualify for.
- Government officers must manually verify a large number of applications.
- Lack of intelligent systems to automatically analyze user eligibility.
- Limited transparency in tracking application status.
- The overall system lacks integration and centralized management of welfare schemes.

CHAPTER 3

METHODS AND TECHNOLOGIES

3.1 Methodology

System Development Approach

The development of the Government Welfare Advisory System follows a structured approach that integrates modern web technologies with machine learning techniques. The system is designed as a full-stack web application where the frontend handles user interaction and the backend manages application logic, database communication, and eligibility prediction. This approach ensures that the system is scalable, efficient, and capable of handling multiple users with different roles such as citizens, officers, and administrators.

The system uses a centralized database to store information related to users, welfare schemes, scheme details, applications, and eligibility history. Machine learning techniques are integrated into the system to analyze user data and predict eligibility scores based on various socio-economic factors. This allows the platform to recommend suitable schemes to citizens automatically, reducing the need for manual searching and improving the accuracy of recommendations.

Web Application Development

The system is implemented as a web-based platform, allowing users to access it through standard web browsers without requiring additional software installation. Web technologies enable the development of responsive and interactive interfaces that make it easier for users to navigate the platform and application tracking.

Machine Learning for Eligibility Prediction

Machine learning techniques are used to analyze user information and determine the probability of eligibility for different welfare schemes. The model

considers factors such as age, income, occupation, gender, and social category to calculate an eligibility score. This predictive approach helps the system recommend schemes that best match the user's profile.

Decision Tree models are often preferred for such applications because they provide transparent and explainable predictions. This is important in government systems where decisions related to welfare benefits must be understandable and auditable.

3.2 Technologies Used

Frontend Technologies

- React.js – Used to build an interactive and responsive user interface.
- TypeScript – Provides type safety and improves the reliability of frontend code.
- Tailwind CSS – Used for styling and creating a modern, responsive design.
- Vite – A fast development tool used for building and running the frontend application.

Backend Technologies

- Spring Boot – Used to develop the backend REST APIs and manage application logic.
- Java – Primary programming language used for backend development.
- Spring Data JPA – Used to simplify database operations and manage entity relationships.

Database Technology

- MySQL – Used to store structured data such as users, schemes, applications, and eligibility records.

Machine Learning Technologies

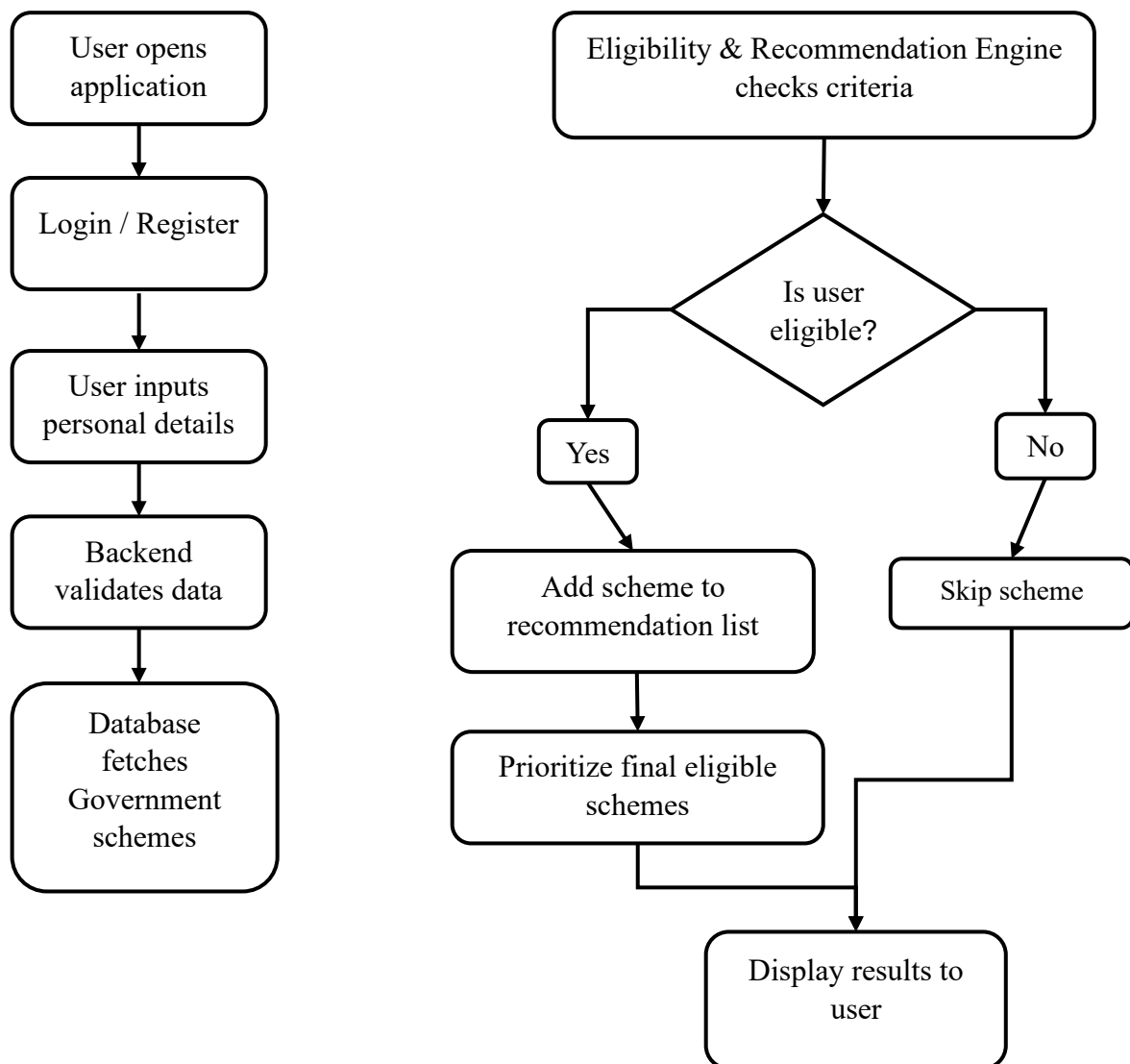
- Python – Used to develop the machine learning model.
- Scikit-learn – Machine learning library
- Flask/FastAPI – Used to expose the machine learning model as an API

Development Tools

- Git – Used for version control and project collaboration.

Flow Chart

Figure 3.1 Flowchart



CHAPTER 4

SYSTEM SPECIFICATIONS

The Government Welfare Advisory System requires minimal hardware as it is a web-based platform. The software requirements include the tools, languages, and frameworks needed to build and run the system.

4.1 Hardware Requirements

Table 4.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 20 GB free disk space
Display	Monitor with at least 1366 × 768 resolution
Network	Stable internet connection

4.2 Software Requirements

Table 4.2 Software Requirements

Category	Software / Technology
Operating System	Windows / Linux / macOS
Frontend	React.js, TypeScript, Tailwind CSS
Backend	Java, Maven, Spring Boot
Database	MySQL
Machine Learning	Python, Scikit-learn
Development Tools	VS Code / IntelliJ IDEA, Git, Postman

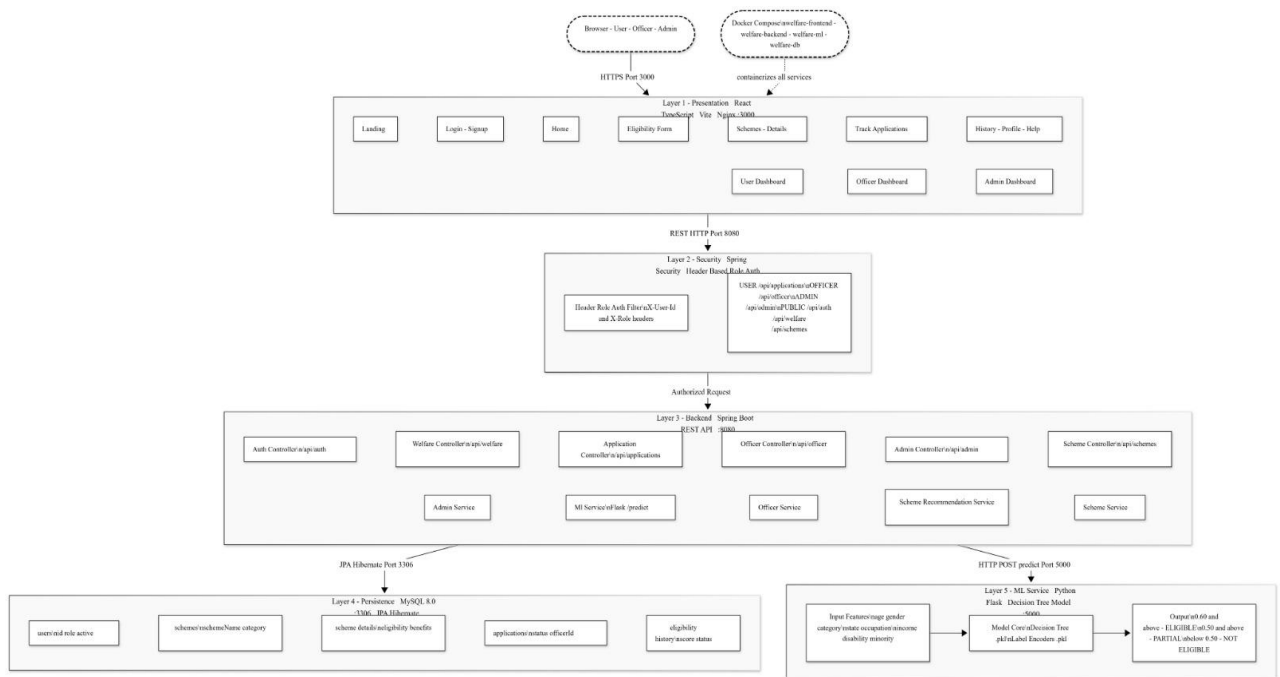
CHAPTER 5

SYSTEM DESIGN

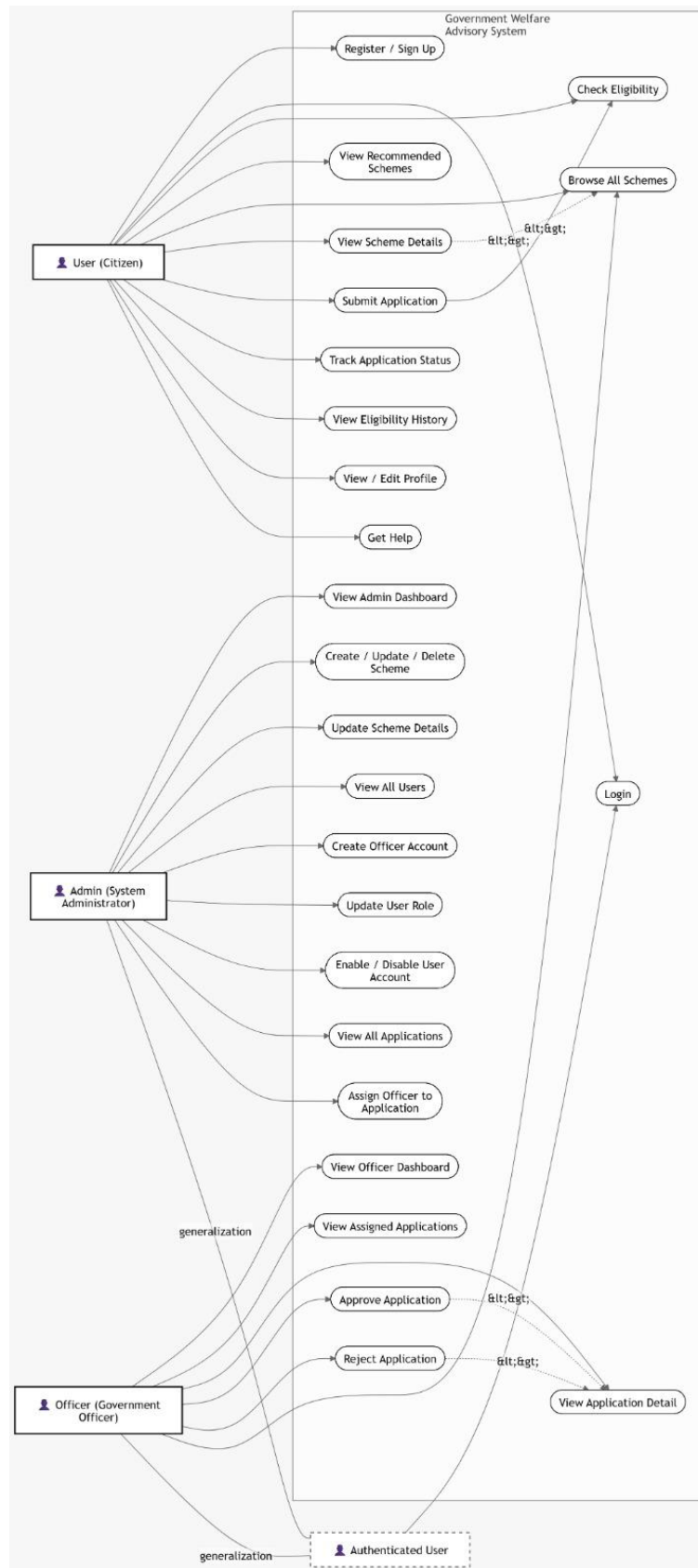
The **Government Welfare Advisory System** is designed as a full-stack web application that integrates frontend, backend, database, and machine learning components to provide a centralized platform for welfare scheme discovery and application management. The system architecture ensures efficient communication between different modules so that users can easily access services such as eligibility checking, scheme recommendation, and application tracking.

The system follows a **client-server architecture**. The frontend interface allows users to interact with the system through a web browser, while the backend processes user requests, performs business logic, and communicates with the database. The machine learning module is integrated with the backend through an API to analyze user data and generate eligibility predictions.

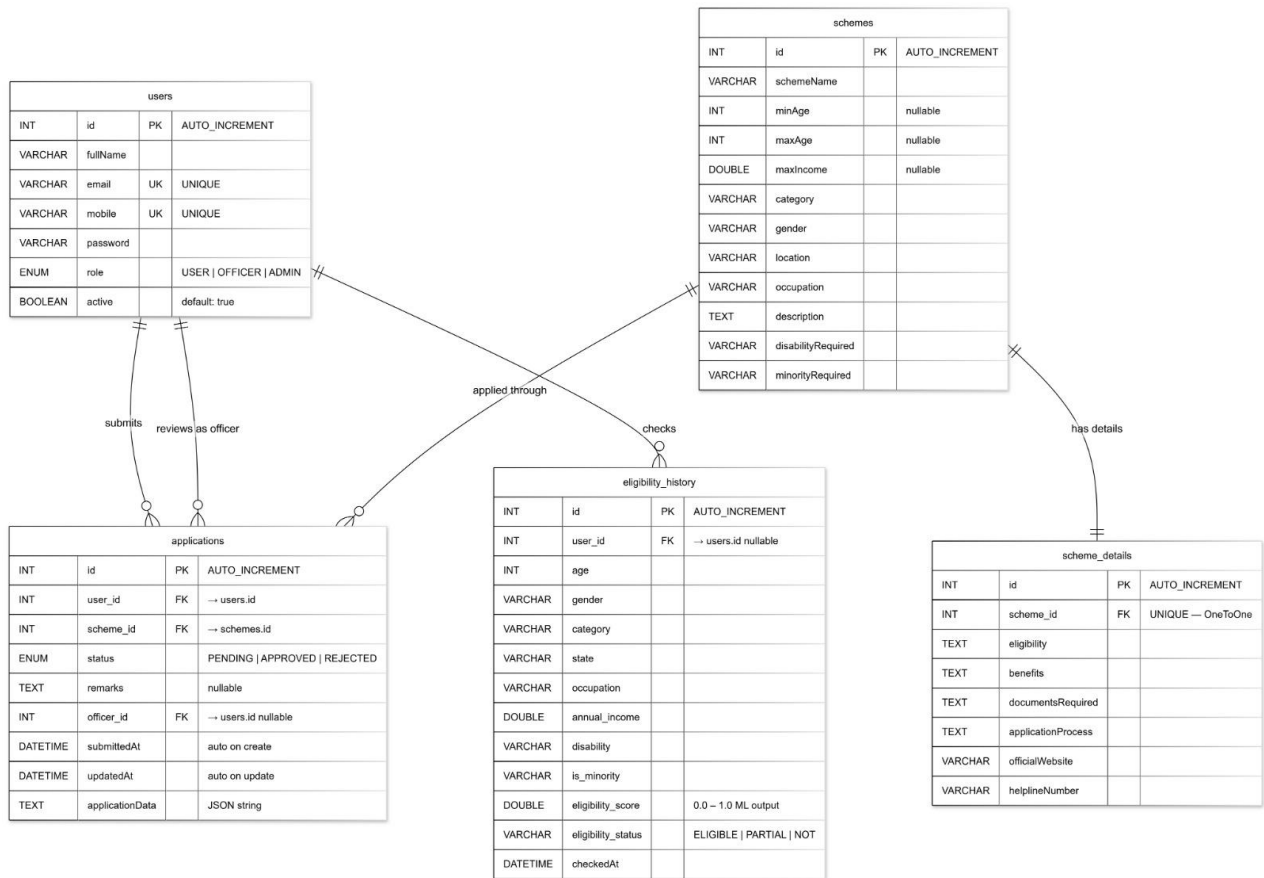
5.1 Architecture Diagram



5.2 Use Case Diagram



5.3 Entity Diagram



CHAPTER 6

IMPLEMENTATION AND EXPERIMENTATION RESULT

6.1 System Implementation

The Government Welfare Advisory System is implemented as a full-stack web application that integrates frontend, backend, database, and machine learning components. The system is designed to provide a centralized platform where citizens can explore welfare schemes, check eligibility, apply for schemes, and track the progress of their applications. The implementation focuses on creating a user-friendly interface, efficient backend services, and an intelligent eligibility prediction model.

The frontend of the system is developed using React.js with TypeScript, which provides a responsive and interactive user interface. Tailwind CSS is used for styling to create a clean and modern design. The frontend communicates with the backend through RESTful APIs to retrieve scheme information, submit user details, and display application data. The system also supports role-based navigation for different users such as citizens, officers, and administrators.

The backend is implemented using Spring Boot, which handles business logic, API development, and communication with the database. The backend manages user authentication, scheme management, eligibility checks, and application processing. A MySQL database is used to store data related to users, welfare schemes, scheme details, applications, and eligibility history. The backend also integrates with a machine learning model through an API to obtain eligibility predictions.

The machine learning component is developed using Python and the Scikit-learn library. The model analyzes user attributes such as age, income, occupation, gender, and social category to generate an eligibility score. The model is exposed

through an API using Flask or FastAPI, allowing the backend to send user data and receive prediction results. This integration enables the system to recommend suitable welfare schemes based on user profiles.

6.2 System Modules Implemented

The system is divided into several modules to manage different functionalities efficiently.

1. User Module

This module allows citizens to register, log in, and manage their profiles. Users can provide personal and socio-economic information that will be used to check eligibility and recommend suitable welfare schemes.

2. Eligibility Checking Module

This module analyzes user information and sends it to the machine learning model. Based on the prediction results, the system calculates an eligibility score and recommends relevant welfare schemes.

3. Scheme Management Module

Administrators can add, update, or remove welfare schemes from the system. They can also manage scheme details such as eligibility conditions, benefits, and application deadlines.

4. Application Management Module

Citizens can apply for welfare schemes through the platform. Officers can review submitted applications and approve or reject them based on verification results.

5. Dashboard and Monitoring Module

The system provides dashboards for different users. Citizens can track their applications, officers can review pending applications, and administrators can monitor system activities.

6.3 Experimental Results

The implemented system was tested to evaluate its performance and functionality. Several test scenarios were conducted to verify that the system correctly recommends welfare schemes and processes user applications efficiently.

During testing, users entered different profiles with varying income levels, occupations, and demographic attributes. The machine learning model successfully analyzed these inputs and generated eligibility scores that were used to recommend relevant welfare schemes. The system demonstrated the ability to quickly process user data and provide scheme suggestions in real time.

The application submission and tracking features were also tested to ensure proper interaction between the frontend, backend, and database. Officers were able to review applications and update their status, which was immediately reflected in the user dashboard. The role-based access control ensured that each user could only access the functionalities assigned to their role.

The experimental results show that the system effectively simplifies the process of discovering welfare schemes and improves the efficiency of application management. By integrating machine learning with a centralized web platform, the system provides a more intelligent and accessible approach to government welfare service delivery.

Fig 6.1 Landing Page

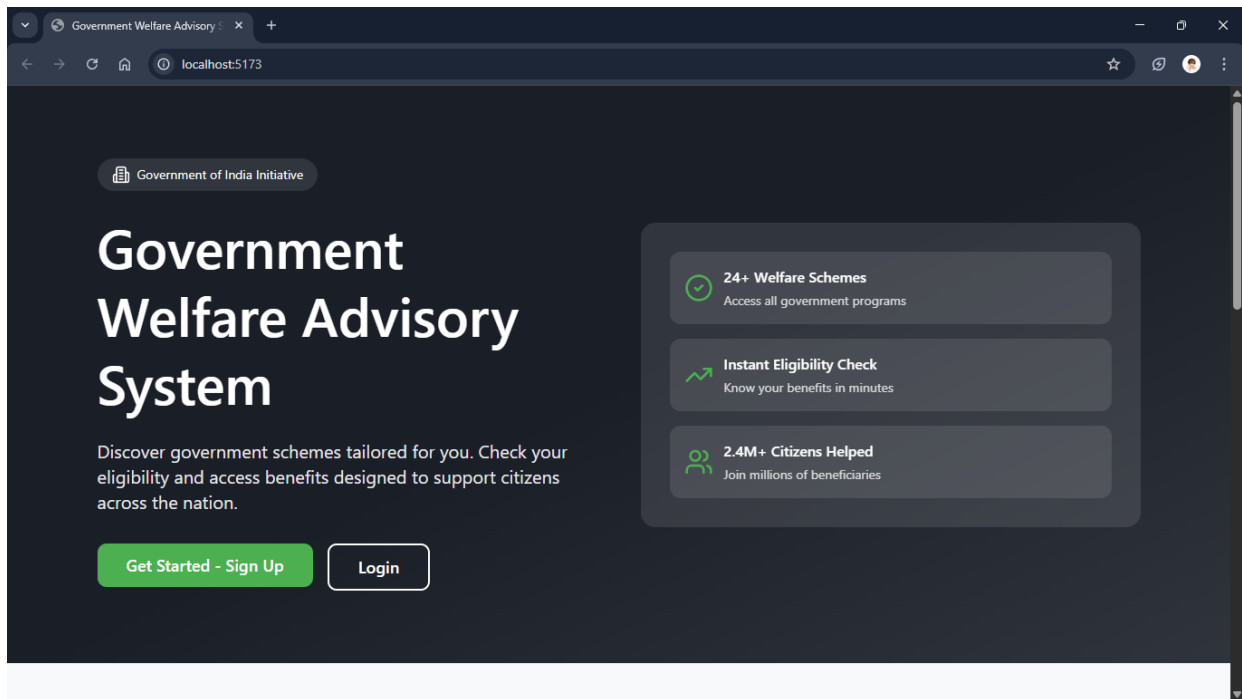


Fig 6.2 Login Page

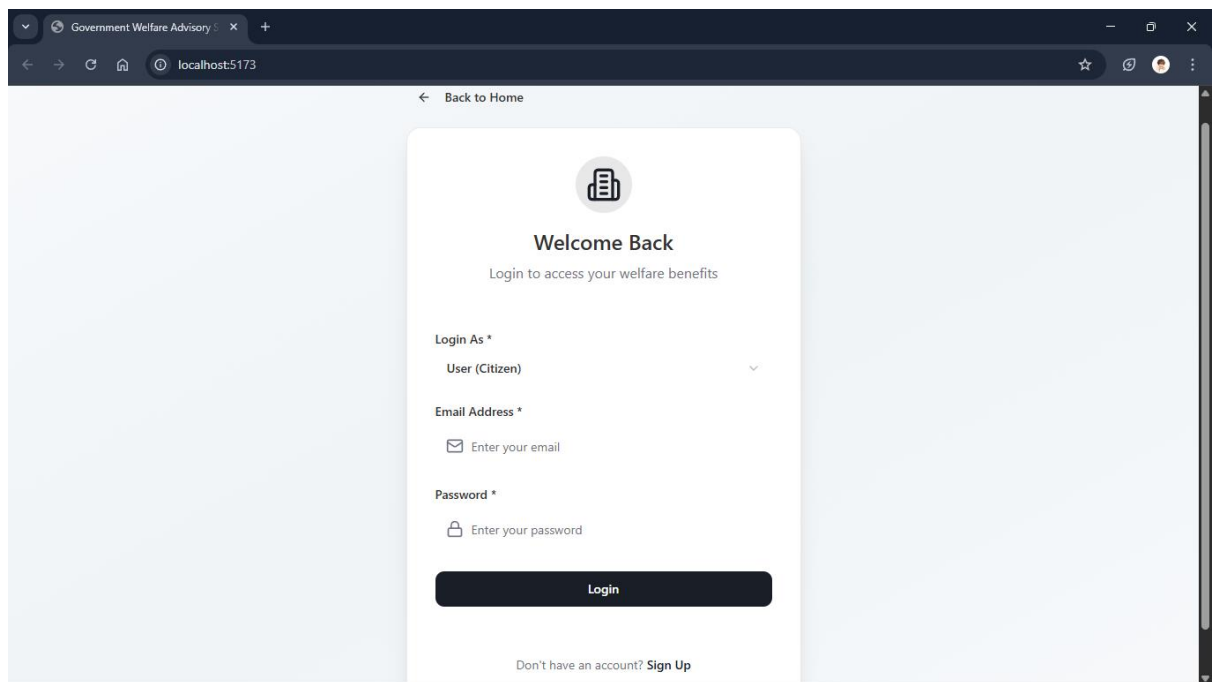


Fig 6.3 User Home Page

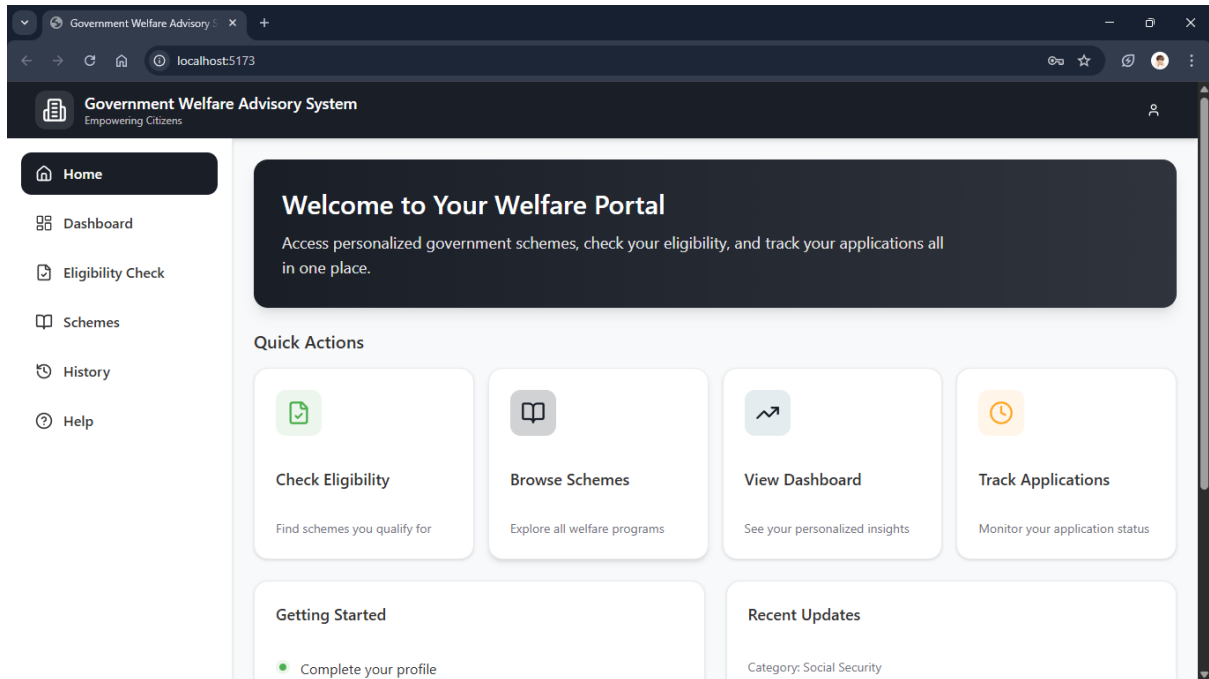


Fig 6.4 Eligibility Checking Page

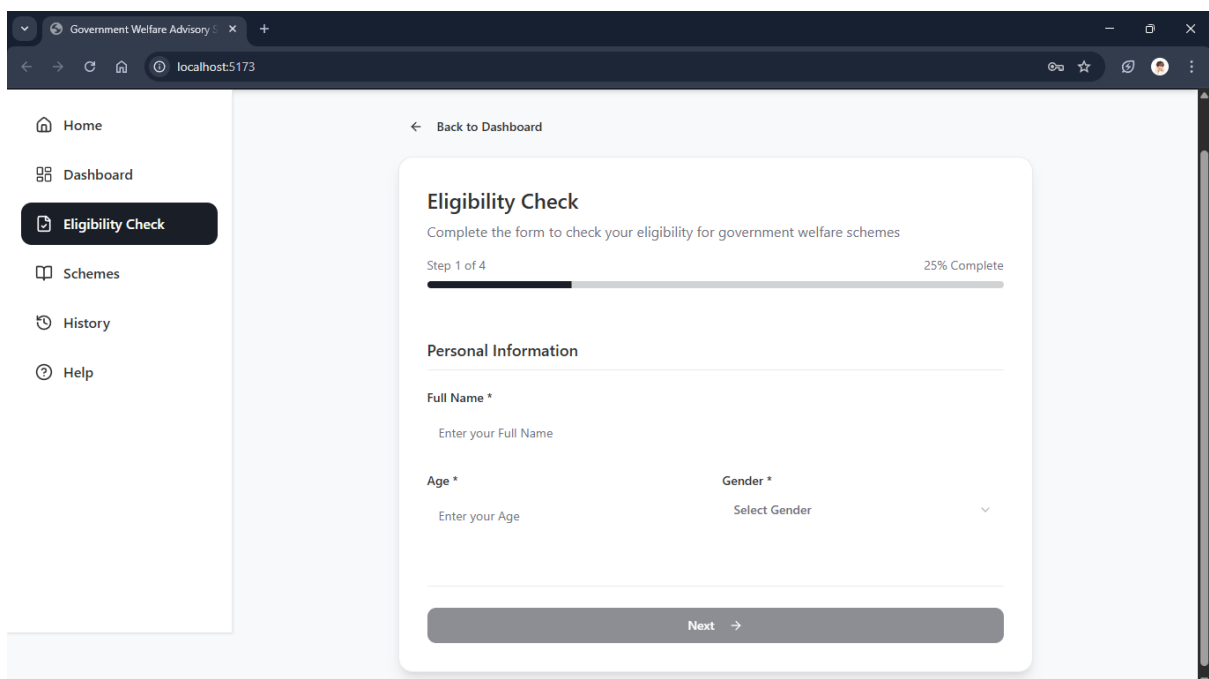


Fig 6.5 Eligibility Check Results

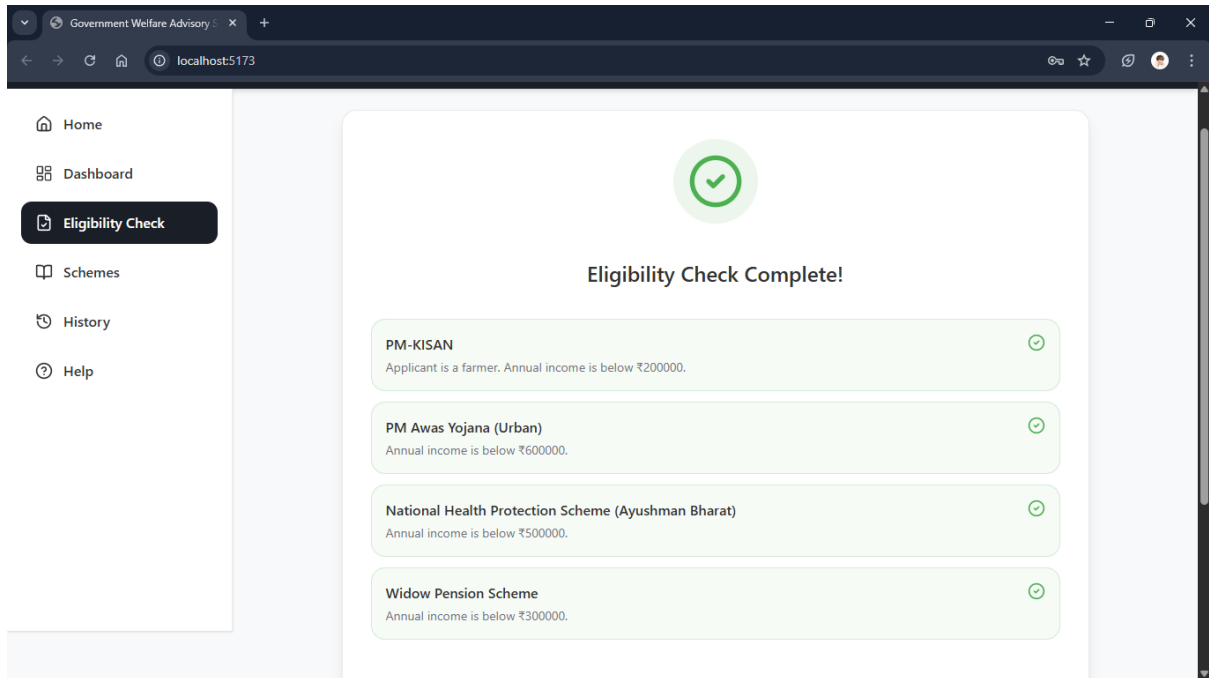


Fig 6.6 Track Applications

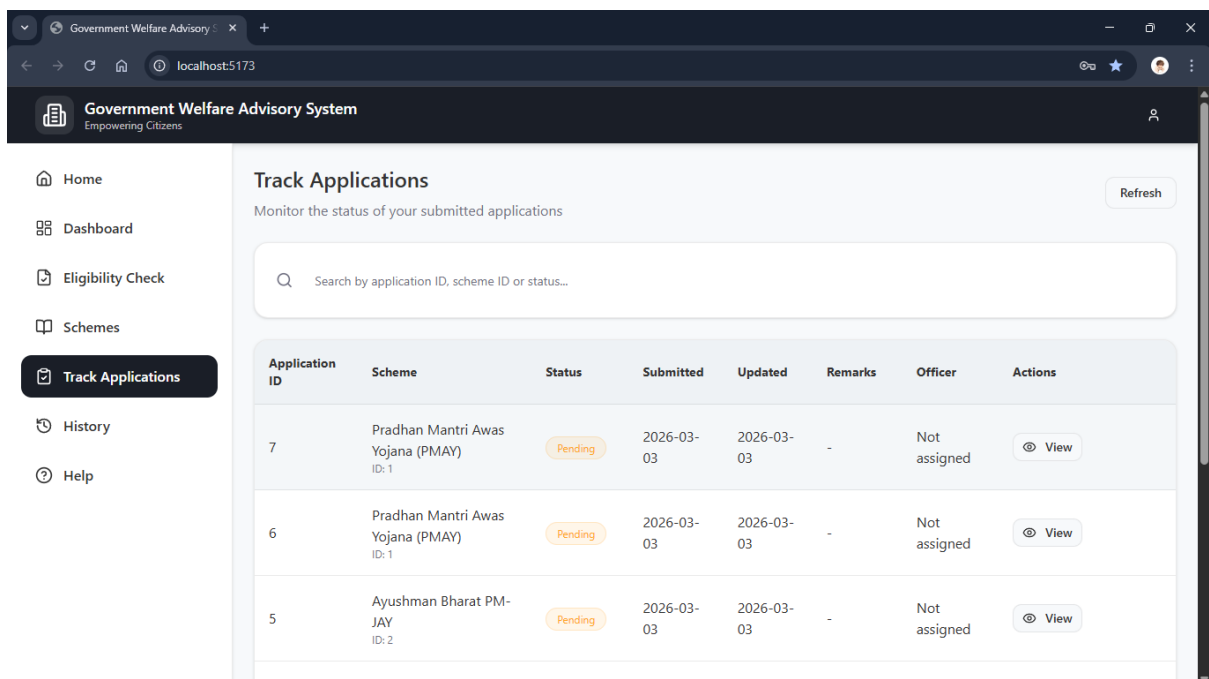


Fig 6.7 Officer Home Page

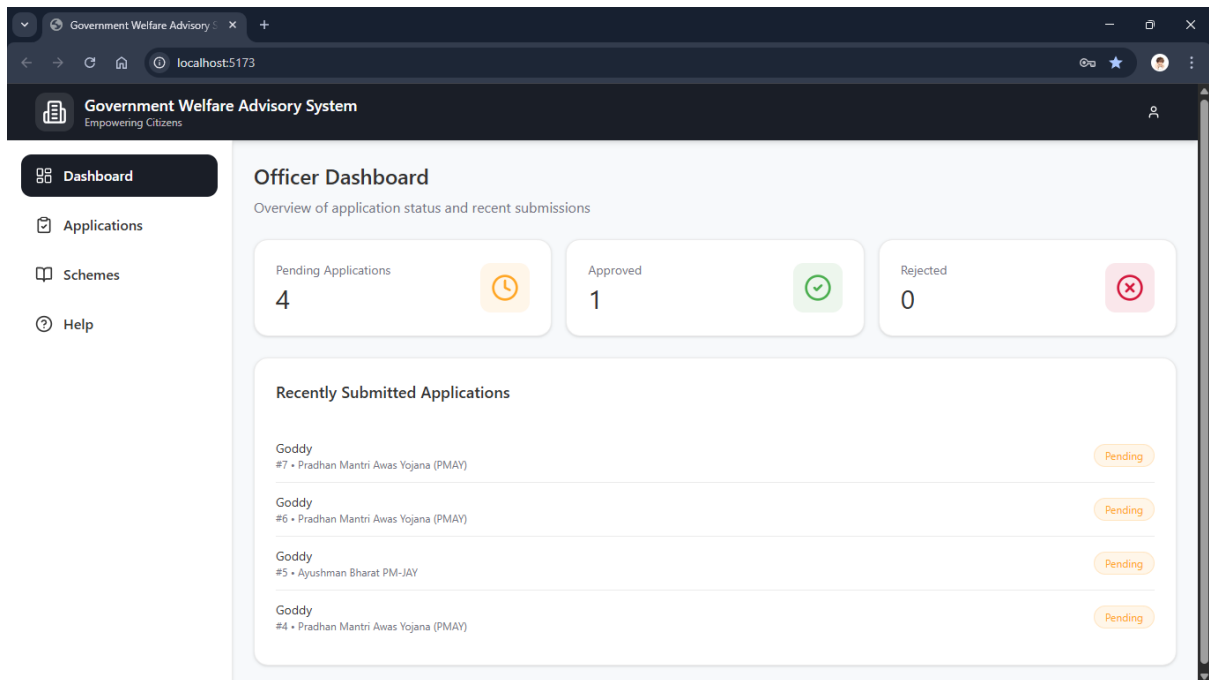


Fig 6.8 Application Review

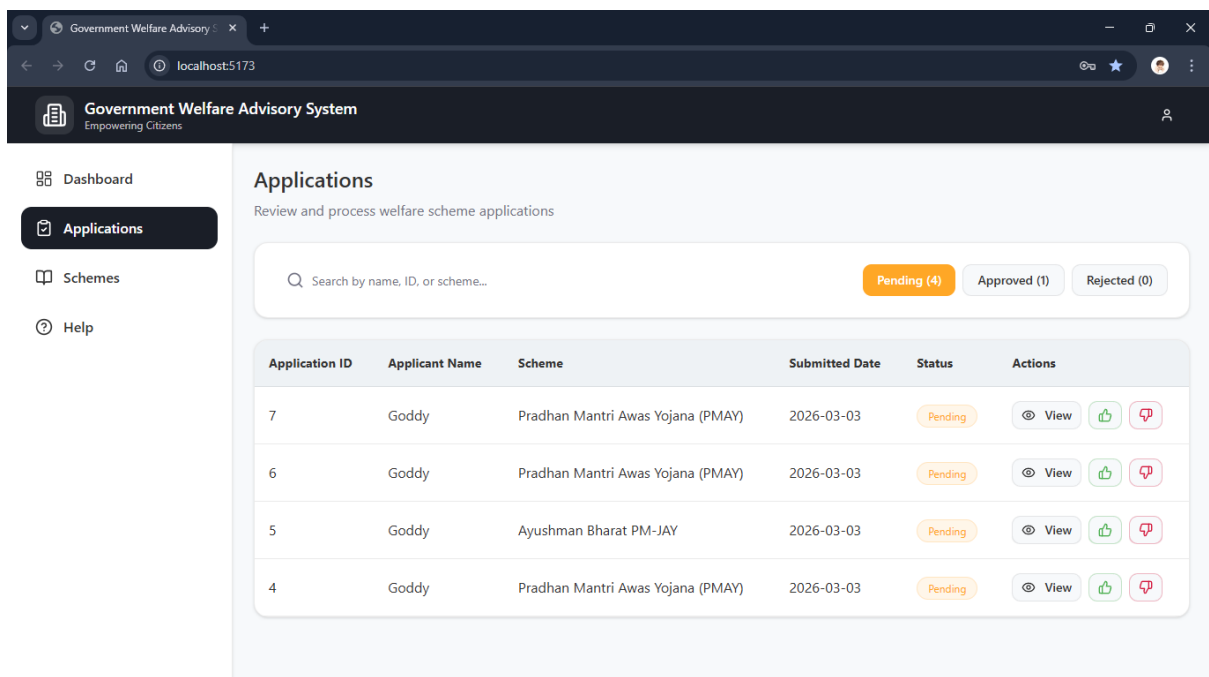


Fig 6.9 Admin Home Page

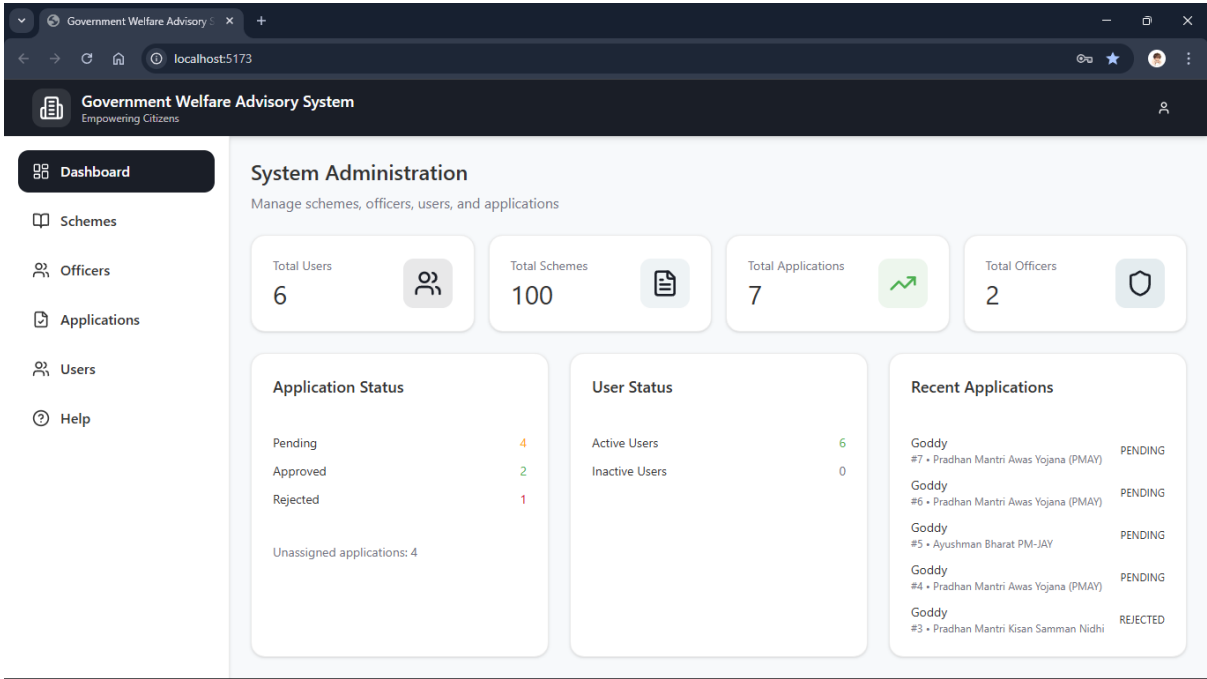
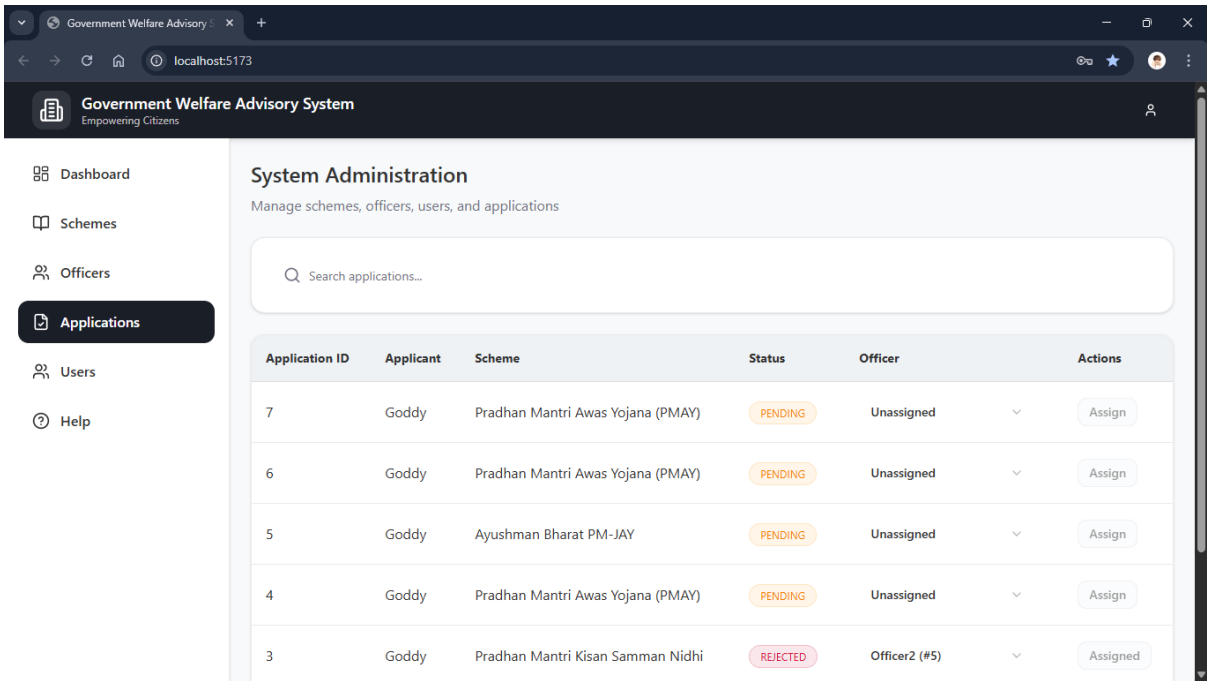


Fig 6.10 Application Assign to Officer



CHAPTER 7

CONCLUSION

The Government Welfare Advisory System was developed to simplify the process of discovering and accessing government welfare schemes. Many citizens face difficulties in identifying schemes they are eligible for due to scattered information and complex eligibility criteria. This project addresses these challenges by providing a centralized digital platform where users can easily check their eligibility, explore suitable schemes, apply for benefits, and track the status of their applications.

The system integrates modern technologies such as React for frontend development, Spring Boot for backend services, MySQL for database management, and machine learning for eligibility prediction. By analyzing user information such as age, income, occupation, and social category, the system provides personalized recommendations for welfare schemes. This reduces the time and effort required for manual searching and helps citizens identify relevant programs more efficiently.

The implementation of role-based access control allows different users such as citizens, officers, and administrators to interact with the system according to their responsibilities. Citizens can apply for schemes and monitor their application progress, while officers can review and process applications. Administrators manage scheme information and system users, ensuring smooth operation of the platform.

The experimental results show that the system successfully improves the process of scheme discovery and application management. The machine learning model effectively predicts eligibility and assists users in finding suitable welfare programs. The integration of web technologies and intelligent data analysis helps create a more accessible and transparent system for welfare scheme distribution.

In conclusion, the Government Welfare Advisory System demonstrates how technology can be used to improve public service delivery and increase awareness of government welfare programs. The system provides a user-friendly, efficient, and scalable solution that helps connect citizens with the benefits intended for them.

7.2 Future Enhancements

- Integration of advanced machine learning models to improve the accuracy of eligibility prediction and welfare scheme recommendations.
- Development of a mobile application to make the platform more accessible for users who prefer accessing services through smartphones.
- Implementation of multilingual support to allow citizens from different regions to use the system in their preferred language.
- Addition of real-time notifications and alerts to inform users about new schemes, application deadlines, and status updates.
- Integration with government databases to automate verification of user information and reduce manual processing.
- Introduction of advanced search features that allow users to find schemes using simple keywords or natural language queries.
- Implementation of data analytics dashboards for administrators to analyze scheme usage, application trends, and beneficiary statistics.
- Enhancement of security features and authentication mechanisms to protect user data and ensure safe access to the platform.

APPENDIX

A1. Source Code

1.ml_service

```
from flask import Flask, request, jsonify

import joblib

import pandas as pd

app = Flask(__name__)

model = joblib.load("../data/processed/welfare_decision_tree_model.pkl")

encoders = joblib.load("../data/processed/encoders.pkl")

FEATURE_ORDER = [

    "age", "gender", "category", "state", "occupation", "annual_income", "disability",

    "is_minority"

]

def safe_label_encode(series, encoder):

    series = series.astype(str)

    return series.apply(

        lambda x: encoder.transform(["Unknown"])[0]

        if x not in encoder.classes_

        else encoder.transform([x])[0]

    )
```

```

@app.route("/predict", methods=["POST"])

def predict_score():

    data = request.json

    df = pd.DataFrame([data])

    for col in FEATURE_ORDER:

        if col not in df:

            df[col] = "Unknown"

    for col, encoder in encoders.items():

        if col in df:

            df[col] = safe_label_encode(df[col], encoder)

    df = df.loc[:, FEATURE_ORDER]

    score = model.predict_proba(df)[0][1]

    return jsonify({

        "eligibility_score": float(score)

    })

if __name__ == "__main__":

    app.run(port=5000, debug=True)

```

2.Scheme Recommendation Service

```

package com.project.welfare.service;

import java.util.ArrayList;

import java.util.List;

```



```

import org.springframework.beans.factory.annotation.Autowired;

import org.springframework.stereotype.Service;

import com.project.welfare.dto.*;

@Service

public class SchemeRecommendationService {

    @Autowired

    private SchemeRepository schemeRepository;

    public List<SchemeRecommendationDto> recommendSchemesWithReasons(

        WelfareRequestDto user,

        String eligibilityStatus) {

        if (eligibilityStatus.equals("NOT_ELIGIBLE")) {

            return List.of();}

        List<SchemeListDto> allSchemes = schemeRepository.findAllForList();

        List<SchemeRecommendationDto> result = new ArrayList<>();

        for (SchemeListDto scheme : allSchemes) {

            Integer minAge = scheme.getMinAge();

            Integer maxAge = scheme.getMaxAge();

            Double maxIncome = scheme.getMaxIncome();

            boolean ageMatch = (minAge == null || minAge <= 0 || user.getAge() >=

minAge);

            if (maxAge != null && maxAge > 0) {

                ageMatch = ageMatch && user.getAge() <= maxAge;}

```

```

        boolean incomeMatch = (maxIncome == null || maxIncome <= 0 ||
user.getAnnual_income() <= maxIncome);

        boolean categoryMatch = matchesField(scheme.getCategory(),
user.getCategory());

        boolean genderMatch = matchesField(scheme.getGender(),
user.getGender());

        boolean locationMatch =

            matchesField(scheme.getLocation(), user.getState()) ||

            isIndiaWide(scheme.getLocation());

        boolean occupationMatch = matchesField(scheme.getOccupation(),
user.getOccupation());

        boolean disabilityMatch =
matchesField(scheme.getDisabilityRequired(), user.getDisability());

        boolean minorityMatch = matchesField(scheme.getMinorityRequired(),
user.getIs_minority());

        if (ageMatch && incomeMatch && categoryMatch && genderMatch
&& locationMatch

            && occupationMatch && disabilityMatch && minorityMatch) {

            StringBuilder reason = new StringBuilder();

            if (minAge != null && minAge >= 60) {

                reason.append("Applicant age is above
").append(minAge).append(". ");}

            if ("Farmer".equalsIgnoreCase(scheme.getOccupation())) {

                reason.append("Applicant is a farmer. ");}

```

```

        if (maxIncome != null && maxIncome > 0) {

            reason.append("Annual income is below ")

                .append(maxIncome.intValue())

                .append(". ");}

        if ("Yes".equalsIgnoreCase(scheme.getDisabilityRequired())) {

            reason.append("Applicant has a disability. ");}

        if ("Yes".equalsIgnoreCase(scheme.getMinorityRequired())) {

            reason.append("Applicant belongs to a minority group. ");}

        result.add(

            new SchemeRecommendationDto(

                scheme.getId(),

                scheme.getSchemeName(),

                reason.toString().trim()));} }

    return result;}

private boolean matchesField(String schemeValue, String userValue) {

    if (schemeValue == null || schemeValue.isBlank()) {

        return true;}

    if (userValue == null || userValue.isBlank()) {

        return false;}

    String normalizedSchemeValue = schemeValue.trim();

    String normalizedUserValue = userValue.trim();

    return normalizedSchemeValue.equalsIgnoreCase("ANY")

```

```

|| normalizedSchemeValue.equalsIgnoreCase(normalizedUserValue);}

private boolean isIndiaWide(String location) {

    return location != null && location.trim().equalsIgnoreCase("India");}}

```

A2 Github Link

Full Complete Code – <https://github.com/Karthi247/Government-Welfare-Advisory-System.git>

A3. API Endpoints

Table 6.1 Api Endpoints

API Endpoint	Method	Description
/api/auth/signup	POST	Register a new user
/api/auth/login	POST	User login authentication
/api/welfare/check-eligibility	POST	Check user eligibility for welfare schemes
/api/welfare/history	GET	View eligibility check history
/api/applications	POST	Submit application for a welfare scheme
/api/applications/user/{userId}	GET	Get applications submitted by a user
/api/officer/applications/pending	GET	View pending applications for officer review
/api/officer/application/{id}/approve	PUT	Approve a welfare application
/api/officer/application/{id}/reject	PUT	Reject a welfare application
/api/admin/schemes	GET	Retrieve all welfare schemes

/api/admin/schemes	POST	Add a new welfare scheme
/api/admin/schemes/{id}	PUT	Update scheme information
/api/admin/schemes/{id}	DELETE	Delete a welfare scheme

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